ANN vs. Quantum-Inspired MPS Signals

A reproducible evaluation inside a FinRL PPO trading agent

Author: Aarav Nagar

Evaluation: Out-of-sample 2019-2023

Prepared: July 24, 2026

Scope: Classical tensor-network simulation; no quantum hardware

Main finding: The MPS predictor had slightly better prediction error and directional accuracy, but its PPO agent did not outperform the matched ANN agent.

Executive summary

Professor Xiao-Yang Liu suggested testing whether a quantum-inspired representation improves sequential decisions when injected into FinRL. I compared Base FinRL, ANN-signal, and QINN-MPS-signal PPO agents under identical dates, assets, costs, policy settings, and seeds.

- Mean Sharpe: ANN 0.735, QINN-MPS 0.694, Base FinRL 0.559.
- MPS trailed ANN in all three paired seeds; the paired annualized return difference was -0.92 percentage points.
- A 20-day block-bootstrap 95% interval was [-4.63, +2.63] percentage points, so the difference is not established.
- Mean annualized turnover was 27.32% for ANN and 24.14% for MPS; transaction costs reconcile to executed notional at 0.10%.

1. Research question and protocol

Research question: Does a parameter-matched quantum-inspired MPS prediction signal improve FinRL trading performance or robustness relative to an ANN signal?

Conditions

Condition	State supplied to PPO	Dimension
Base FinRL	Cash, prices, holdings, 10 indicators/asset	181
ANN signal	Base state + one frozen ANN signal/asset	196
QINN-MPS	Base state + one frozen MPS signal/asset	196

Chronology and controls

- 2013-2017: fit the ANN and MPS encoders.
- 2018: encoder validation and early stopping.
- 2013-2018: PPO policy training.
- 2019-2023: locked out-of-sample evaluation.

The same 15 stocks, 13 encoder inputs, next-day target, optimizer family, 0.10% buy/sell cost, PPO network, 5,000-step budget, and seeds 0, 1, and 2 were used.

Parameter-matched encoders

The ANN uses a 13-16-8-1 tanh network. The MPS uses a trigonometric local feature map and bond dimension 4. Each has exactly 369 trainable parameters. The MPS runs classically and does not imply quantum advantage.

Exact signal entry

Base state = 1 cash + 15 prices + 15 holdings + (10 indicators x 15 assets) = 181 values.

Signal state = base state + one standardized, clipped, frozen encoder prediction per asset = 196 values. The 15 predictions are appended as the final per-asset indicator block.

2. Prediction and portfolio results

Prediction metrics on untouched 2019-2023 targets

Encoder	Params	MSE	MAE	Direction	IC
ANN	369	1.5101	0.7994	50.25%	+0.0119
QINN-MPS	369	1.5034	0.7967	51.06%	-0.0060

MPS had slightly lower error and higher directional accuracy, while its information coefficient was slightly negative.

Mean portfolio results

Condition	Return	Sharpe	Max DD	Turnover	Cost
Base FinRL	73.24%	0.559	-40.46%	26.43%	$1,375
ANN signal	104.66%	0.735	-27.55%	27.32%	$1,430
QINN-MPS signal	94.27%	0.694	-26.79%	24.14%	$1,254
Equal-weight	218.25%	1.115	-28.54%	20.06%	$1,000

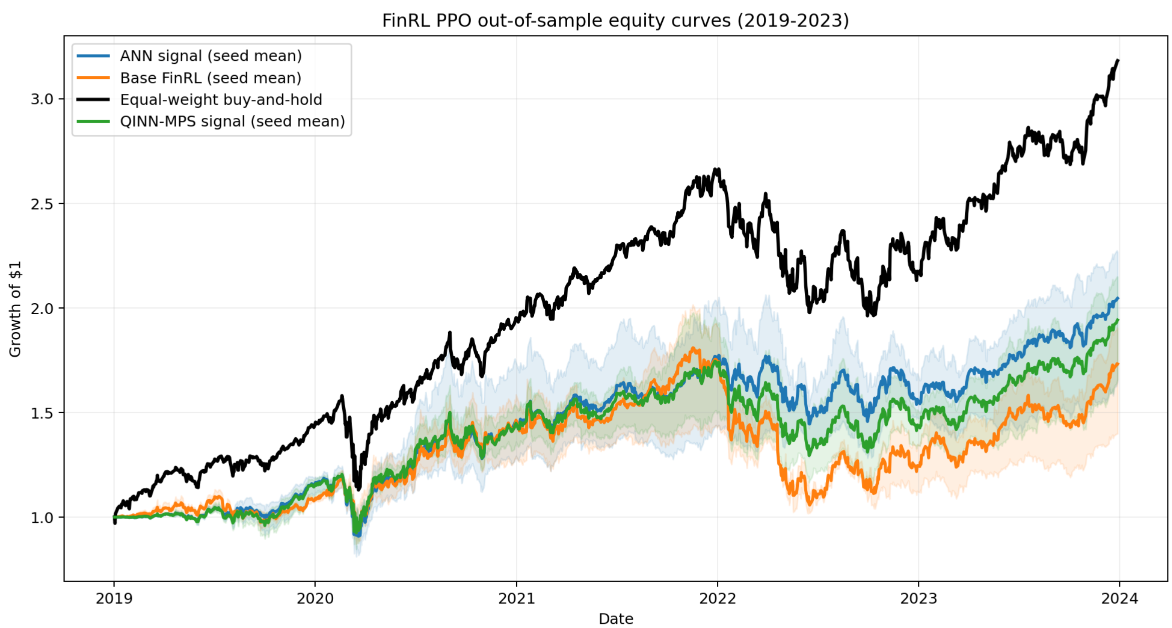

Figure 1. Out-of-sample equity curves; shading spans PPO seeds.

3. Seeds, uncertainty, turnover, and costs

Seed-level Sharpe

Seed	Base	ANN	MPS	MPS - ANN
0	0.436	0.790	0.759	-0.032
1	0.633	0.812	0.726	-0.086
2	0.608	0.602	0.596	-0.006

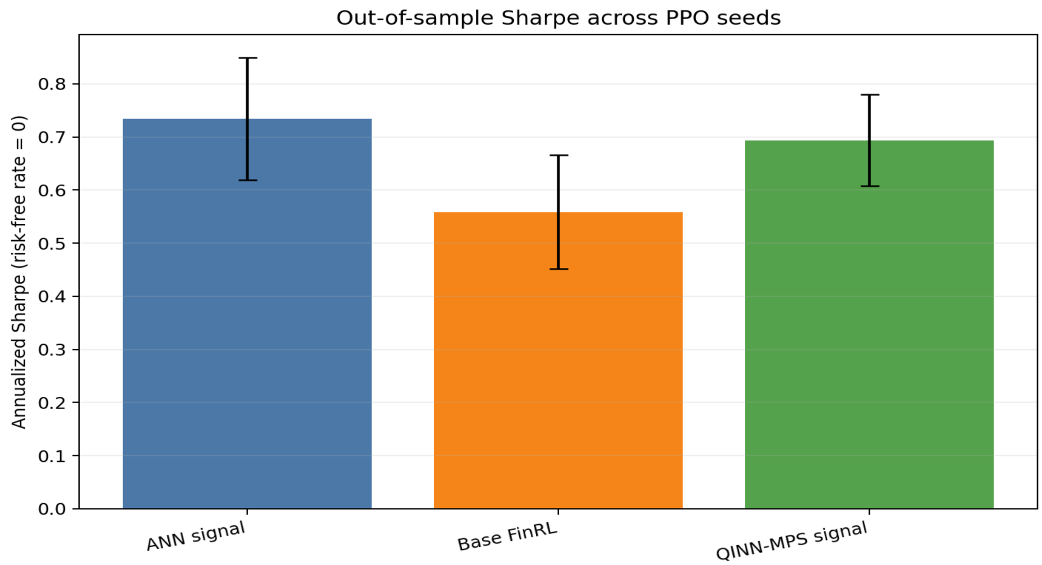

Figure 2. Mean Sharpe and seed-to-seed standard deviation.

Paired uncertainty

A moving 20-day block bootstrap with 2,000 samples estimated the annualized MPS-minus-ANN mean-return difference at -0.92%, with a 95% interval of [-4.63%, 2.63%]. The interval includes zero.

Turnover and transaction costs

Daily gross turnover equals executed share notional divided by beginning-of-step portfolio value. Mean annualized turnover was 27.32% for ANN, 24.14% for MPS, and 26.43% for Base FinRL.

At a fixed 0.10% fee, five-year mean costs were $1,429.65 for ANN, $1,253.61 for MPS, and $1,375.38 for Base FinRL. Costs reconcile exactly to gross executed notional x 0.001.

4. Market-period robustness

Year	Base	ANN	MPS	Equal-weight
2019	8.00%	15.69%	14.85%	43.52%
2020	35.48%	25.85%	26.28%	36.13%
2021	18.36%	20.48%	18.70%	34.88%
2022	-27.71%	-9.24%	-15.79%	-17.47%
2023	38.40%	30.72%	34.55%	46.34%

Calendar years avoid selecting favorable regimes after seeing results. MPS beat ANN in 2020 and 2023; ANN beat MPS in 2019, 2021, and 2022. No stable MPS advantage appears.

Trading was concentrated early. Mean annualized turnover fell from about 103% for ANN and 100% for MPS in 2019 to below 1% for both by 2023.

Interpretation

Supported conclusion: For this dataset, parameter budget, PPO configuration, three seeds, and historical split, a quantum-inspired MPS signal did not improve FinRL trading performance over an equally sized ANN signal.

The result does not generalize to all tensor networks. It demonstrates the prediction-decision gap: MPS slightly improved some prediction metrics, but ANN produced higher Sharpe in every paired trading seed.

Limitations

- One chronological split and a 15-stock Nasdaq subset.
- Three seeds and a scoped 5,000-step PPO budget.
- Classical MPS simulation; no quantum hardware.
- Fixed 0.10% fee but no spread, slippage, impact, taxes, or borrow costs.
- Calendar years do not replace rolling-window evaluation.

5. Recommended ATL/FinRL continuation

The strongest next step is to strengthen the evaluation before adding a tensor-network policy head.

1. Increase PPO training to at least 50,000 steps and use 10 or more seeds.
2. Add rolling or expanding-window retraining.
3. Predeclare and compare several MPS bond dimensions.
4. Add exposure, concentration, slippage, and crisis-window diagnostics.
5. Then compare a tensor-network policy head against this external-signal design.

Reproducibility package

- run_experiment.py - complete data, encoder, PPO, and evaluation pipeline.
- test_experiment.py - six focused integrity tests.
- results/ppo_backtest_metrics.csv - per-seed full-period metrics.
- results/equity_curves.csv - daily returns, notional, turnover, and costs.
- results/annual_period_metrics.csv - calendar-year metrics by seed.
- results/condition_seed_summary.csv - exploratory seed intervals.
- results/ann_vs_mps_block_bootstrap.csv - paired uncertainty analysis.
- results/run_manifest.json - source pins, checksums, protocol, and definitions.

References

1. Biamonte, J. et al. Quantum Machine Learning. Nature 549, 195-202 (2017).
2. Liu, X.-Y. and Fang, Y. Quantum Tensor Networks for Variational Reinforcement Learning. NeurIPS 2020 workshop.
3. NeurIPS 2020 and 2021 Quantum Tensor Networks in Machine Learning workshop accepted-paper pages.
4. FinRL Contest 2025 Task 1: FinRL-DeepSeek for Stock Trading.
5. Wang, K. et al. FinRL Contests: Benchmarking Data-driven Financial Reinforcement Learning Agents. arXiv:2504.02281.

Full URLs and an expanded reference list are included in research_report.md.